Sahaj Bhadja

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Summary

Full-Stack Software Engineer with 2+ years building **secure React applications**, **Python APIs**, cloud services, AI workflows, access controls, automated tests, and observable production systems.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Advanced NLP, Database Systems, Web Technologies, Analysis of Algorithms

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Operating Systems, Object-Oriented Programming, Data Retrieval

Experience

USC Dornsife **September 2025 – Present**
Software Engineer *Los Angeles, CA*

- Translated stakeholder workflows into **fault-tolerant LLM orchestration** with FastAPI workers, explicit job states, timeouts, failure isolation, and monitoring, processing 1K+ concurrent AI jobs at 3x throughput under load.
- Built multi-tenant integrations across Azure SQL, PostgreSQL, SQL Server, and cloud storage, processing 5K+ monthly jobs with **row-level isolation, validation, CI/CD, and 99%+ reliability** across regulated production environments.

Grad8 **June 2025 – August 2025**
Software Engineer Intern *Los Angeles, CA*

- Built **React and Flask AI workflows** for uploads, grading, history, and feedback, supporting 150+ instructors and 10K+ daily requests while reducing turnaround 60% through production releases for daily use.
- Implemented secure file validation, concurrent processing, REST APIs, model routing, and safe fallbacks; coordinated releases through **automated testing, CI/CD, monitoring, and production issue investigation** across multiple environments.

Aasma Technology Solutions **August 2023 – July 2024**
Software Developer Engineer - (SecureSync) *India*

- Designed object-oriented microservices and large-scale API integrations, improving response times 30% while collaborating with Microsoft and GitHub teams through **architecture reviews, Agile planning, and release coordination** with stakeholders.
- Automated **security-conscious testing and deployment evidence** with 20+ PyTest and Mockito suites, Jenkins pipelines, VM builds, and cross-platform validation, increasing pipeline efficiency 68% and reducing technical debt 20%.

Cyber Security and Data Mining **May 2023 – August 2023**
Data Analyst - Research Fellowship *India*

- Built a **malicious-URL detection pipeline** over 650K+ URLs using 21 behavioral indicators, statistical feature engineering, Random Forest, and deep learning models for phishing and malware analysis at scale.
- Evaluated RNN, LSTM, Bi-LSTM, GRU, CNN, and classical models through reproducible experiments, achieving **96%+ detection accuracy** while documenting failure patterns and actionable security insights for repeatable analysis.

Aasma Technology Solutions **May 2023 – August 2023**
Full Stack Developer Intern - (VibrantChem) *India*

- Built eight microservices for authentication, permissions, resources, billing, and notifications, using **REST APIs and event-driven WebSockets** to synchronize 2M+ records across distributed locations through secure services at scale.
- Developed React and TypeScript dashboards, then supported releases through **Docker images, Jenkins CI/CD, Sentry monitoring, audit-friendly logging, and troubleshooting** across eight customer-facing production services with reliable incident response.

Skills

Languages: TypeScript, JavaScript, Python, Java, SQL
Full Stack: React, FastAPI, Flask, Node.js, REST APIs, Microservices, WebSockets, Event-Driven Systems
AI & Security: LLM APIs, RAG, Tool Calling, Structured Outputs, Embeddings, Authentication, Authorization, RBAC, Input Validation, Audit Logging
Data & Cloud: PostgreSQL, SQL Server, MongoDB, Azure, Amazon Web Services, GCP, Docker, Kubernetes
Operations: CI/CD, Jenkins, GitHub Actions, Git, Unit Testing, Integration Testing, Code Reviews, Debugging, Monitoring

Projects

SecureMCP Gateway | *Python, FastAPI, MCP, PostgreSQL, OAuth, Docker, Kubernetes*

- Designed a **secure MCP integration gateway** exposing security tools through typed schemas, structured outputs, scoped authorization, attributable audit logs, explicit approval gates, bounded retries, timeouts, and reliable error handling.

SecOps Copilot | *TypeScript, React, Python, RAG, LLM APIs, Vector Search, Monitoring*

- Built an **AI-assisted security operations workspace** for alert enrichment, evidence gathering, log summarization, and triage with retrieval, tool-access controls, iteration limits, human checkpoints, regression tests, and safe fallbacks.

Big-Brain | *LLaMA-2, FAISS, Pinecone, PostgreSQL, GCP, REST APIs* | Live

- Built a **secured RAG application** with LLaMA-2, transformer embeddings, FAISS search over 50K+ documents, Clerk authentication, and PostgreSQL persistence, cutting retrieval latency 40% at 95% query accuracy.